

Index Number:

Candidate Name: _____

CT Group: 07S_____

HWA CHONG INSTITUTION



JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY 9747

PAPER 2

11 Sept 2008

2 hrs

INSTRUCTIONS TO CANDIDATES

There are **4** question booklets (**I** to **IV**) to this paper. Write your **name**, **CT** and **Index Number** on the cover pages of the booklets in the spaces provided. Write in blue or black pen.

SECTION A

This section contains **7** questions. Answer **ALL** questions.

Write your answers in the spaces provided on the question paper.

SECTION B

Answer **ONE** question.

Your answer to **Section B** must be in continuous prose, where appropriate.

Write your answers on the separate answer paper provided.

At the end of the examination,

1. write the number of the **Section B** question you have answered in the opposite grid. **A NIL return is required.**
2. submit all question booklets and answer sheets to **Section B** separately.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the at the end of each question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
6	
7	
Total	/ 100

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

Fig. 1.1 and **Fig. 1.2** show the structures of glycogen and triglyceride respectively. Both glycogen and triglyceride are metabolic fuels.

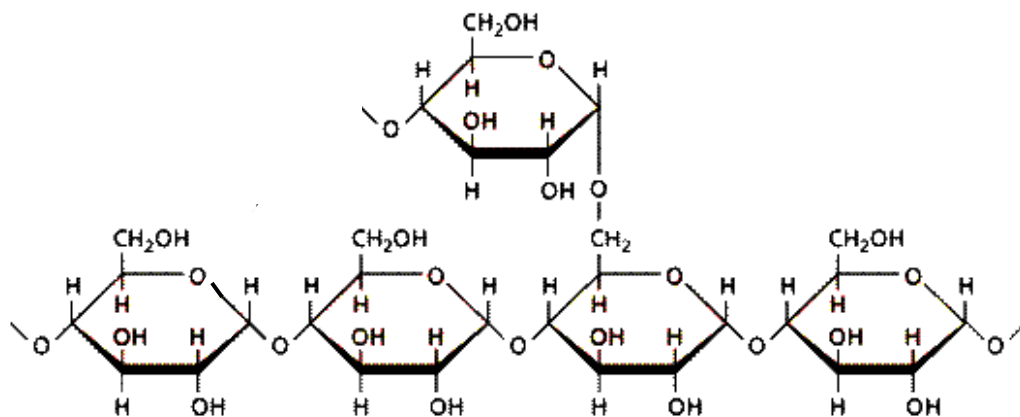


Fig. 1.1

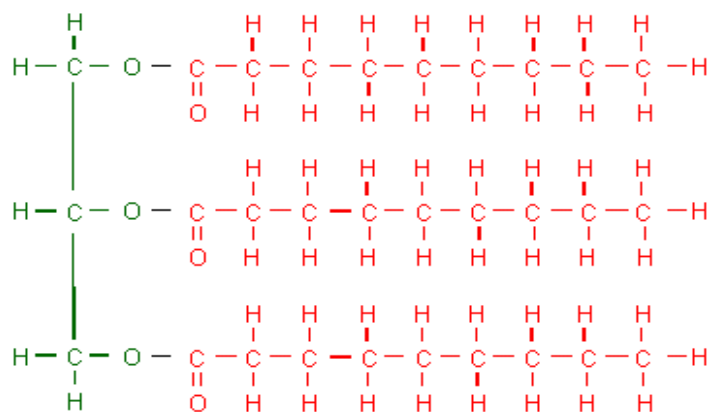


Fig. 1.2

With reference to **Fig. 1.1** and **Fig. 1.2**,

- (a) state one similarity and one difference in the structure of glycogen and triglyceride. [2]

Similarity: _____

Difference: _____



Fig. 1.3 shows some steps involved in glycolysis and the Krebs cycle. Some ATP is made directly. Hydrogen is also released and this can result in the production of more ATP.

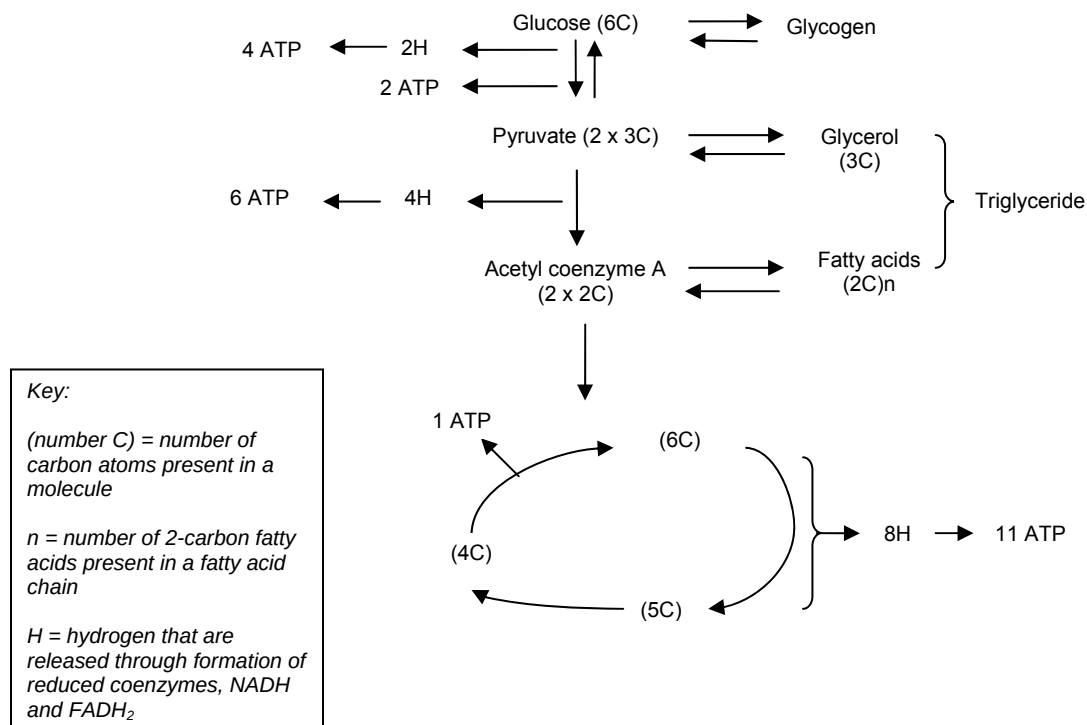


Fig. 1.3

(b) Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP. [2]

Similar to glycogen, triglycerides can be respired. The first step is to hydrolyse each triglyceride molecule. Each fatty acid formed is broken down into acetyl CoA molecules. The acetyl CoA molecules then enter the Krebs cycle.

Using the information from **Fig. 1.3**,

(c)(i) calculate the number of ATP molecules produced from the complete oxidation of a 16-carbon fatty acid. Show your answer in the space provided below. [2]

(ii) suggest why fatty acids can only be respired under aerobic conditions. [2]

Glucose is considered as one of the most important metabolites in the blood. **Fig. 1.4** shows how blood glucose levels are strictly regulated in the human body.

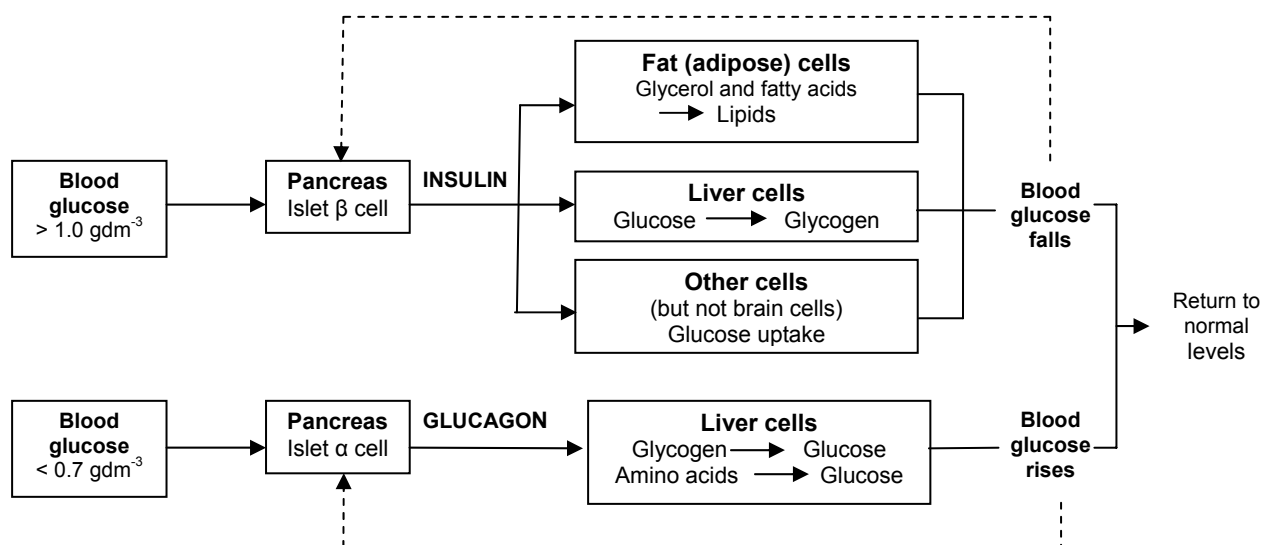


Fig. 1.4

(d)(i) Explain two important principles of homeostasis that are illustrated in **Fig. 1.4**. [3]

(ii) Suggest why glucose is considered as one of the most important metabolites. [1]

[Total: 12 marks]

QUESTION 2

Fig. 2.1 shows cultured fibroblasts of the Indian barking deer, *Muntiacus muntjac*, undergoing different stages of mitosis, **J** to **Q**.

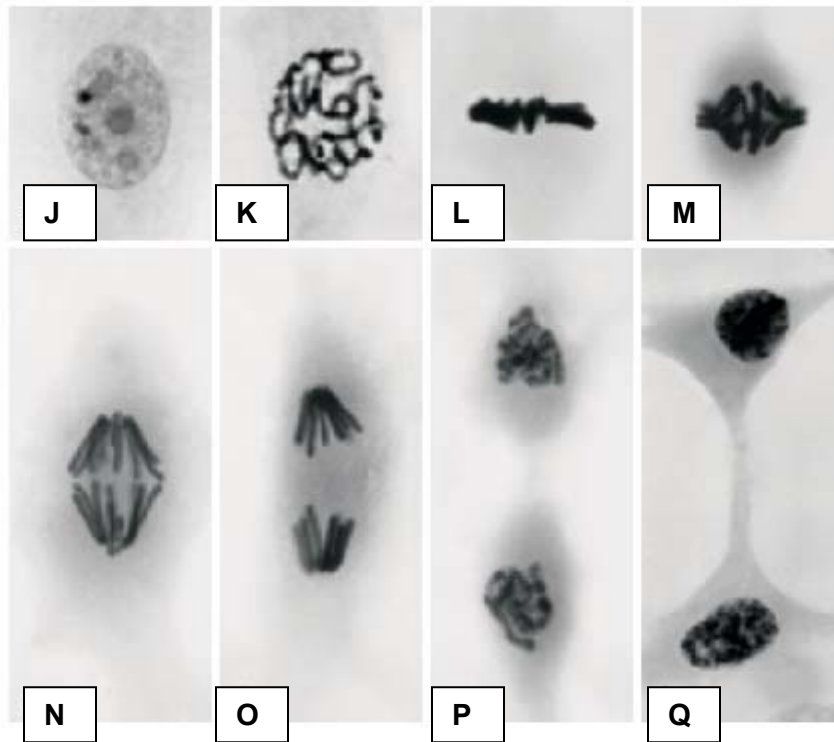


Fig. 2.1

(a)(i) With reference to **Fig. 2.1**, identify stages **K**, **L**, **O** and **Q**. [2]

K: _____

L: _____

O: _____

Q: _____

(ii) Describe the behaviour of chromosomes at stages **N** and **P**. [2]

N: _____

P: _____

Studies of cell cultures of these fibroblasts have revealed that progress between phases of the cell cycle is normally controlled by 'checkpoints'. If these fail, then tumours can develop.

- (b) The M checkpoint occurs during mitosis of the cell cycle. Explain the significance of this checkpoint. [2]

The G₁ checkpoint functions to regulate the entry of cells into S phase, hence controlling rate of cell proliferation. This checkpoint can be regulated by a few factors, including proto-oncogenes and tumour suppressor genes. **Table 2.1** shows the results of cell counts from normal tissues of the Indian barking deer, where the stages of mitosis are scored separately from cells in interphase. **Table 2.2** shows the results obtained when tumour cells from the same deer are cultured.

Stage	Number of cells	% of cells at each stage
Interphase	924	87.3
Prophase	46	4.4
Metaphase	48	4.5
Anaphase	8	0.8
Telophase	32	3.0

Table 2.1

Stage	Number of cells	% of cells at each stage
Interphase	60	42.9
Prophase	17	12.1
Metaphase	32	22.9
Anaphase	16	11.4
Telophase	15	10.7

Table 2.2



(c)(i) With reference to **Table 2.1**, describe the differences in percentage distribution of cells at each stage. [2]

(ii) Using information from **Table 2.1** and **Table 2.2**, explain the effect of oncogenes on cell division. [3]

(iii) Suggest why it is important to use percentage values when comparing data from the normal and tumour cell cultures. [1]

[Total: 12 marks]

BOOKLET II

SECTION A: STRUCTURED QUESTIONS (Continued)

QUESTION 3

- (a) Describe how binary fission in bacteria brings about genetic stability. [2]

While binary fission ensures genetic stability, genetic variation can still occur in bacteria through genetic exchange and recombination. A process that brings about variation is conjugation.

- (b) Describe how genetic variation is brought about through conjugation. [2]

In conjugation, a Hfr (High frequency of recombination) cell functions as a donor. DNA replication is initiated at a specific point on the integrated F factor DNA, and from that point a single strand of the F factor DNA moves into the F⁻ partner, dragging along adjacent chromosomal DNA in a specific order. Using the principles of high frequency of recombination, scientists are able to generate maps of genes found on bacterial chromosomes. One such study is carried out between two *Escherichia coli* strains shown in **Table 3.1**. Strain **A** is the donor strain.

Strain	Phenotype
A	<i>str</i> ^S , <i>azi</i> ^r , <i>ton</i> ^r , <i>glu</i> ⁺ , <i>lac</i> ⁺
B	<i>str</i> ^r , <i>azi</i> ^S , <i>ton</i> ^S , <i>glu</i> ⁻ , <i>lac</i> ⁻

The following abbreviations are used for the phenotypes:
str – resistance (r) or sensitivity (s) to streptomycin, an antibiotic,
azi – resistance (r) or sensitivity (s) to sodium azide,
ton – resistance (r) or sensitivity (s) to bacteriophage T1,
glu – ability (+) or inability (-) to utilize glucose as an energy source, and
lac – ability (+) or inability (-) to utilize lactose as an energy source

Table 3.1



Strains **A** and **B** are mixed. At specific times following the mixing, a fraction of the mixture is removed and put in a blender for some seconds. The bacteria are then plated on specific media plus streptomycin, and the growth is scored as shown in **Fig. 3.1**.

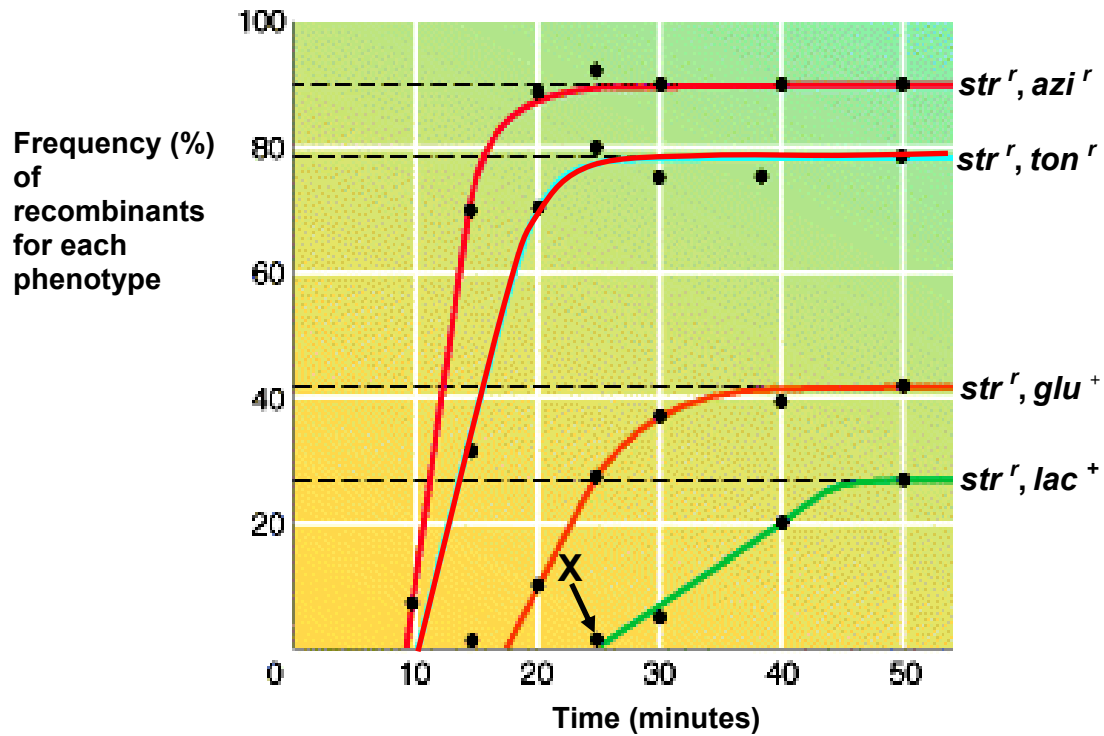


Fig. 3.1

(c) Describe the trend observed in **Fig. 3.1**. [2]

Fig. 3.2 shows a portion of the recipient bacterial chromosome after conjugation. The labelled vertical lines represent the respective gene loci.

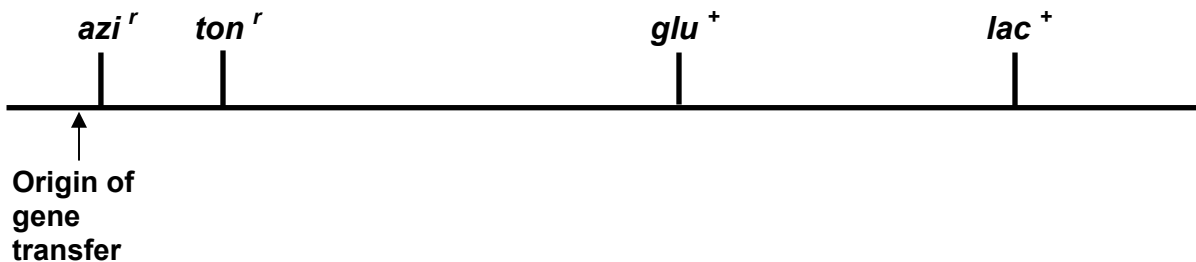


Fig. 3.2

(d) Explain the relative positions of the genes seen in **Fig. 3.2**, using results from **Fig. 3.1**. [2]

The experiment is repeated. At point **X**, the mixture is divided into two tubes with growth medium containing streptomycin and glucose, but lactose is added to one of the tubes.

(e)(i) State any difference there might be in the growth rate between the two tubes. [1]

(ii) Explain your answer to (e)(i). [2]

[Total: 11 marks]

QUESTION 4

In an experiment, chromatin from various chick tissues were isolated and treated with DNase I, an enzyme that degrades free double-stranded DNA in a non-specific manner. After digestion, the DNase I enzyme was removed. Any remaining intact DNA was extracted and mixed with radioactively labelled cDNA probes specific for certain genes, under conditions that favoured nucleic acid hybridisation.

The levels of binding of the labelled cDNA probes were measured. Some of the results are shown in **Table 4.1** below.

Sample	Tissue source of chromatin	DNase I treatment	Radioactive cDNA probe	% binding of radioactive cDNA to remaining DNA after DNase I digestion
1	Developing red blood cells (nucleated)	-	Globin cDNA	94
2	Developing red blood cells (nucleated)	+	Globin cDNA	25
3	Oviduct	+	Globin cDNA	91
4	Developing red blood cells (nucleated)	+	Ovalbumin cDNA	93
5	Oviduct	+	Ovalbumin cDNA	18

Table 4.1

Globin: protein subunit of haemoglobin

Ovalbumin: main protein present in egg white

In animals that lay eggs, the formation of the shelled egg occurs in the oviduct, the passage from the ovaries to the outside of the body.

If the cDNA probe finds sequences to bind to, then the gene concerned is said to be 'protected' from DNase I digestion by chromatin proteins.

(a)(i) Name one such chromatin protein. [1]

(ii) Explain how these proteins protect the gene concerned from DNase I digestion. [2]

With reference to **Table 4.1**,

(b)(i) describe the differences in the levels of binding of the two types of cDNA probe for the developing red blood cells samples. [2]

(ii) state which gene is more actively transcribed in developing red blood cells. [1]

(iii) explain how the experimental results support your answer to **(b)(ii)**. [2]

In the developing chick red blood cell, an embryonic form of globin is first synthesised, while the adult form, β -globin, is only synthesised later in the formation of mature red blood cells. **Fig. 4.1** (not drawn to scale) below shows the organisation of the β -globin gene in the chick.

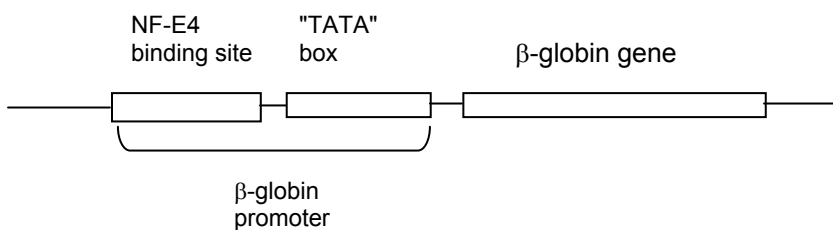


Fig. 4.1

Experiments have shown that the NF-E4 protein is found at higher levels in maturing red blood cells than in embryonic chick red blood cells. NF-E4 protein has a DNA-binding domain.

(c)(i) State the role of the β -globin promoter. [1]

- (ii) With reference to **Fig. 4.1**, explain how β -globin expression in the developing red blood cell may be regulated by regulating levels of NF-E4 within the nucleus. [2]

[Total: 11 marks]

BOOKLET III

SECTION A: STRUCTURED QUESTIONS (Continued)**QUESTION 5**

In an investigation on the tomato genome, pollen grains from the anthers of a pure-bred plant with green leaves and hairy fruit were transferred by a scientist to the stigmas of a pure-bred plant with mottled (green and yellow) leaves and smooth-surfaced fruit. All the F_1 generation had green leaves and smooth fruit.

- (a) Define what is meant by a pure-bred plant. [1]

A test cross was then made between the F_1 plant and a pure-bred plant with mottled leaves and hairy fruit. The phenotypes of 100 offspring of the test cross were recorded and shown in **Table 5.1**.

Green leaves, smooth fruit	Green leaves, hairy fruit	Mottled leaves, smooth fruit	Mottled leaves, hairy fruit
8	46	40	6

Table 5.1

- (b) Using suitable symbols, draw a genetic diagram to explain the results of the test cross. [5]

- (c) Explain the deviation of the observed test cross results from the expected dihybrid test cross ratio. [2]

A second scientist repeated the test cross between the same F₁ plant and pure-bred plant with mottled leaves and hairy fruit. However for this time, he only recorded the number of the 2 more common phenotypes of 100 offspring of the test cross as shown in **Table 5.2**.

Green leaves, hairy fruit	Mottled leaves, smooth fruit
52	44

Table 5.2

The formula for the chi-squared (χ^2) test is given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degrees of freedom	Probability		
	0.10	0.05	0.01
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

- (d)(i) This second scientist claimed that the observed test cross ratio of offspring can be simplified to 1:1. Perform the chi-squared (χ^2) test to verify his claim. [2]



(ii) Suggest why the second scientist made this claim.

[1]

[Total: 11 marks]

QUESTION 6

Fig. 6.1 and **Fig. 6.2** below show the structures of the rhodopsin receptor, the photoreceptor on the retina.

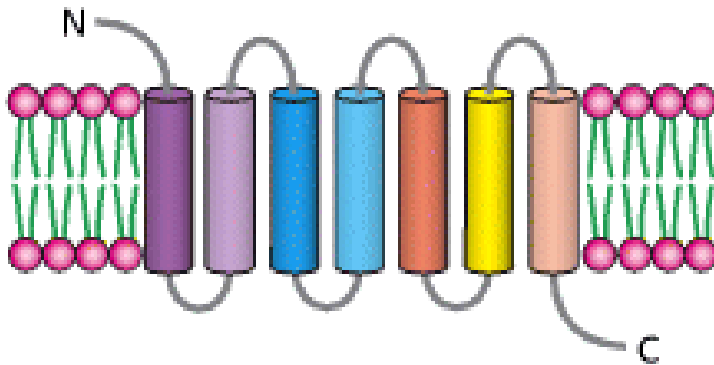


Fig 6.1

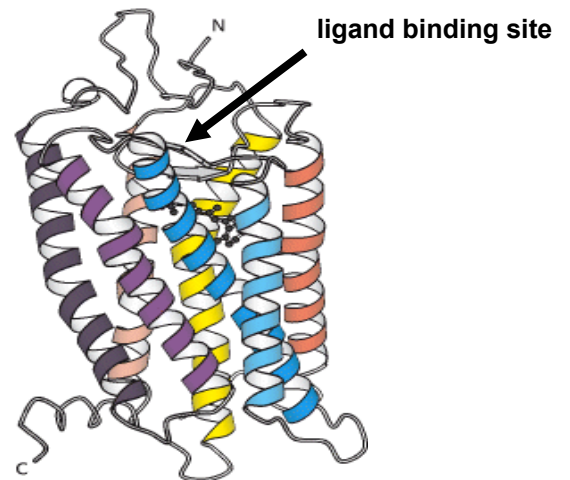


Fig 6.2

(a)(i) State the level of protein structure shown in **Fig. 6.2**. [1]

(ii) With reference to **Fig. 6.1** and **Fig. 6.2**, explain how the structure of the rhodopsin receptor allows it to carry out its function. [2]

Researchers have elucidated the steps of the signalling pathway labelled (A to F) which rhodopsin receptor is involved in. This is shown in **Fig. 6.3**.

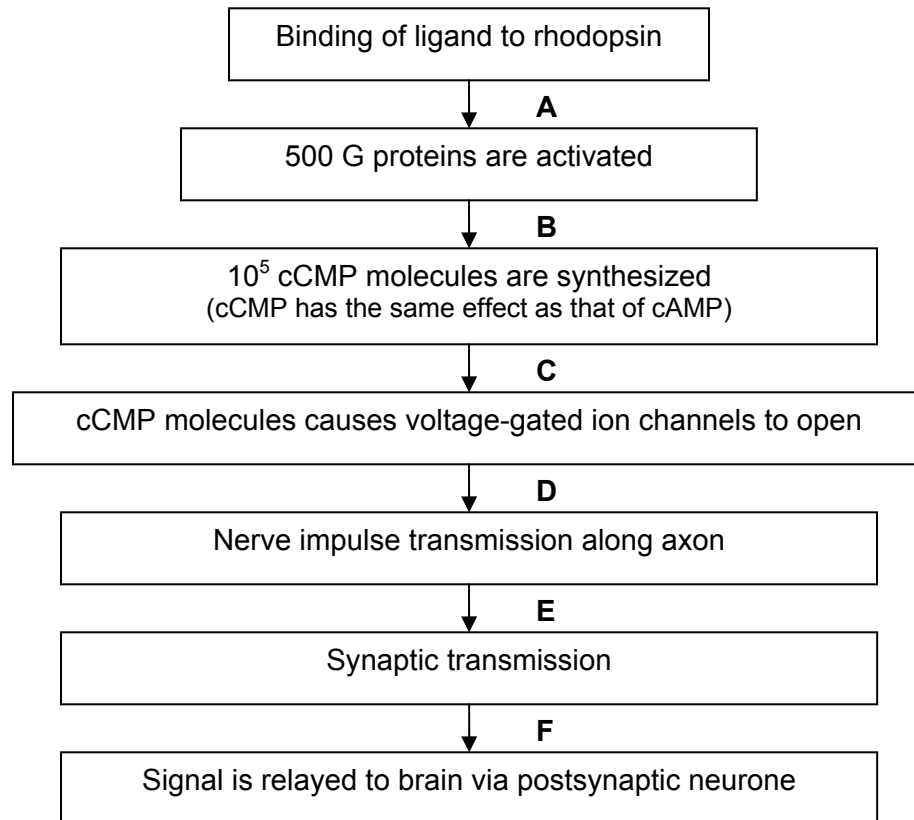


Fig. 6.3

(b)(i) State any two steps in **Fig. 6.3** which result in amplification effect of this signalling pathway. [1]

(ii) Explain the significance of the signal amplification effect. [2]

In a separate experiment, a group of researchers investigated how stimulus intensity affects nerve impulse transmission along the axon and synaptic transmission at the neuromuscular junction. They varied the strength of the stimulus and measured the extent of muscular contraction by recording the number of twitches produced per second. Electrodes were placed at different regions of the neurone to measure the changes in membrane potential generated after stimulation.

The experiment was repeated three times, **E1** to **E3**, using a different stimulus intensity each time. The results of the experiments are shown in **Fig. 6.4** and **Table 6.1**.

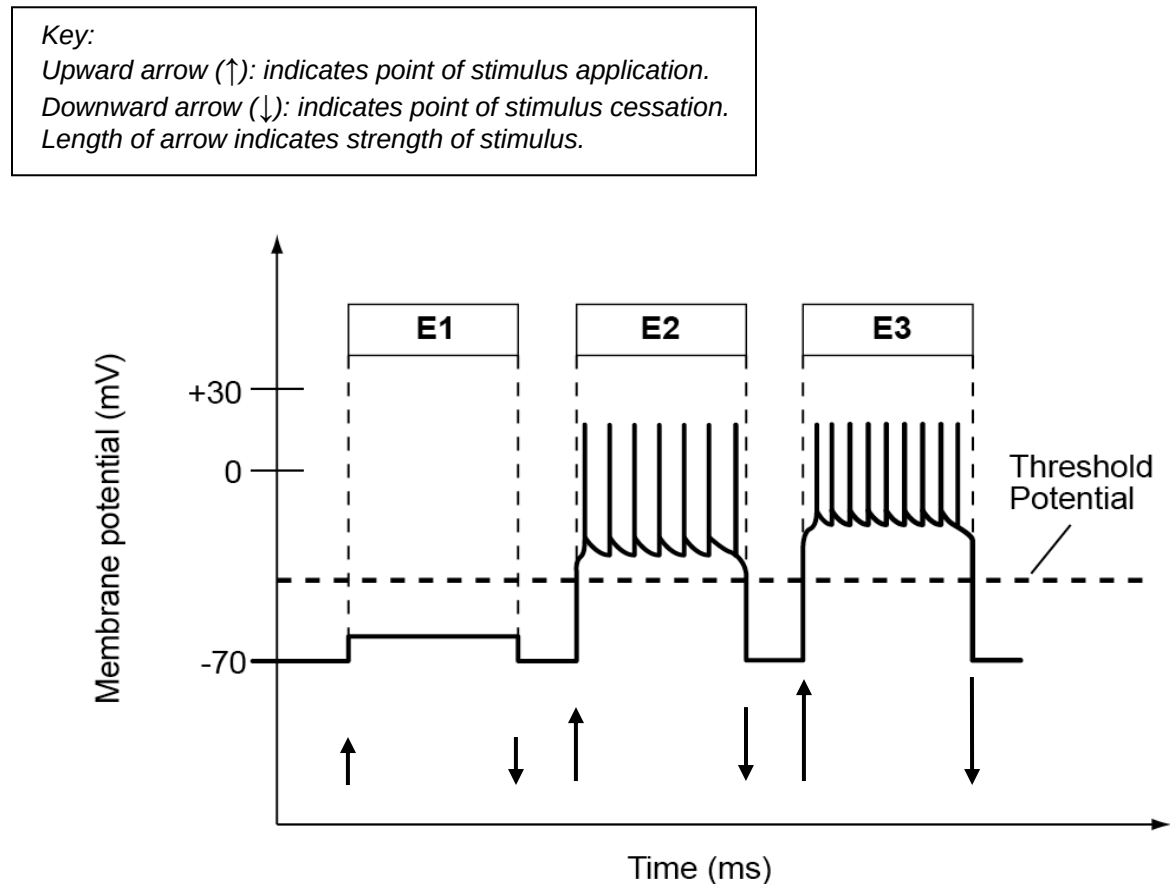


Fig. 6.4

Experiment	Number of twitches produced per second
E1	0
E2	7
E3	11

Table 6.1

(c) Describe the trend observed in **Fig. 6.4** for **E2** and **E3**. [2]

(d)(i) State what you understand by the term 'threshold potential'. [1]

(ii) With reference to the results shown in **Fig. 6.4** and **Table 6.1**, explain how stimulus intensity affects the extent of muscular contraction. [3]

[Total: 12 marks]

BOOKLET IV

SECTION A: STRUCTURED QUESTIONS (Continued)

QUESTION 7

The traditional classification of Vertebrata recognizes seven taxonomic classes, of which only four are shown in **Fig. 7.1**. One phylogenetic classification recognizes 13 clades, of which 8 are shown.

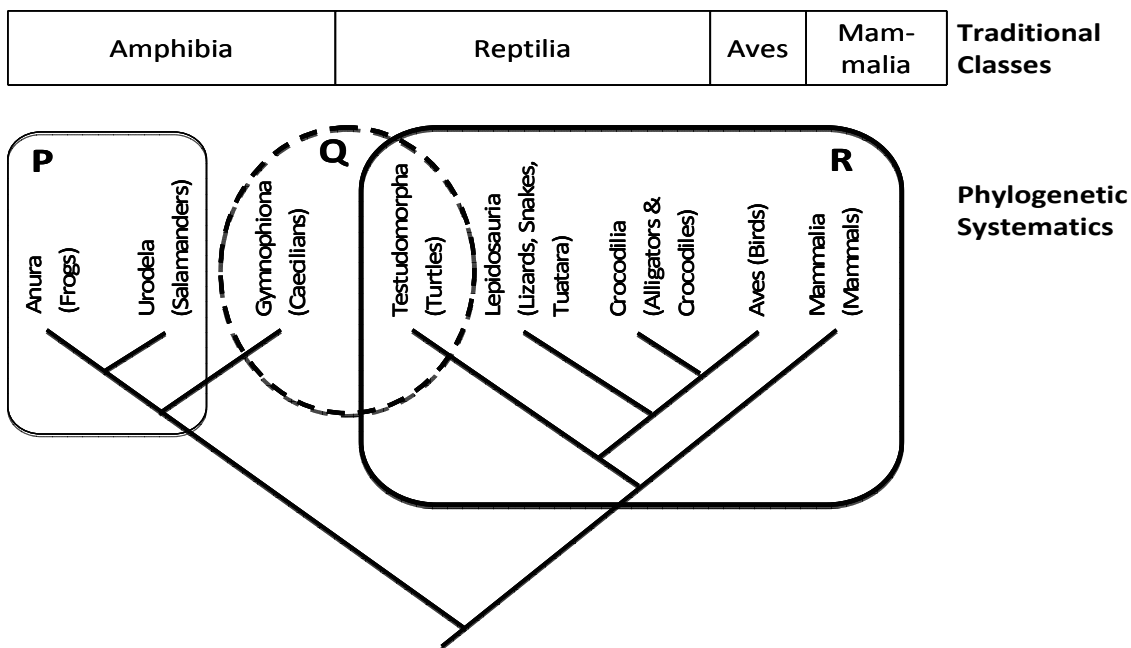


Fig. 7.1

(a)(i) State one difference between traditional classification and phylogenetic systematics. [1]

(ii) P, Q and R show 3 types of groupings. Identify the groupings. [3]

P: _____

Q: _____

R: _____

(iii) Explain how these types of groupings are derived.

[2]

There are several subspecies of the mouse, *Mus musculus*, living throughout the world. A study was conducted to study their spread from an ancestral population. **Fig. 7.2** shows the results of sequencing mitochondrial DNA from 10 mice from geographically diverse locations. Dashes indicate bases that are identical to the top sequence.

England	AAAGCAGAAGAAATAGATGCAAAGAAGAAGAGTTCAA
Japan	--GCAT-----C-----AATT
South Africa	-----C-----TA-----
India	GG-CG--C-----GGG-GG-----CGGG
China	TTT--T-----TT--T-----GGTT
United States	-----C-----T-----
Philippines	TTT--T-----TC--T-----GGTT
New Zealand	-----C-----C-TA-----
Russia	--GCAT-----CC-----A-TT
Yemen	-----C-----

Fig. 7.2

(b)(i) Draw a box in **Fig. 7.2** to include the entire region that most likely codes for an important structure in the mitochondrion. [1]

(ii) Explain your answer to (b)(i). [2]

(iii) With reference to **Fig. 7.2**, identify the country of the subspecies which is most closely related to the mouse found in England. Give a reason for your answer. [2]

[Total: 11 marks]

--- End of Section A ---



SECTION B: FREE RESPONSE QUESTIONS

Answer **ONE** of the following two free-response questions. A **nil return** is required.

Your answers must be in continuous prose, where appropriate.

Answers should be illustrated by large, clearly visible diagrams, where appropriate.

QUESTION 8

- (a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]
- (b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [6]
- (c) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [8]

QUESTION 9

- (a) Describe the structural features of two named nucleic acids involved in protein synthesis. [8]
- (b) Compare the two processes that have to take place for protein synthesis. [8]
- (c) Discuss two named factors that would affect the rate of protein synthesis. [4]

--- End of Section B ---

